



# **National University of Modern Languages**

## **Assignment No 3**

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|----------------------|------------------------------------|
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| <b>Course:</b>       | <b>Computer Graphics</b>           |
| <b>Class:</b>        | <b>BSCS VII-A</b>                  |
| <b>Due Date:</b>     | <b>5<sup>th</sup> -April, 2025</b> |
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## **Task: Be able to create a simple 3D game in Unity 3d and be familiar with the basics of game development.**

### **Scenario:**

Create a Simple 3D game in Unity 3D that should be developed based on following criteria.

1. Scene contains geometry added via the Unity editor
2. Scene contains a camera, and a screenshot of the scene can be captured
3. Scene incorporates at least one texture
4. Scene incorporates at least two different types of light (e.g. a directional light and a point light)
5. Scene includes at least one object imported from another source (e.g. made in Blender, Downloaded online, etc. Anything not created in Unity)

### **Instructions**

1. Make proper documentation of assignment along with screenshots.
2. Make powerPoint presentation of your work along with live demonstration.

### **1.1 Objective**

The goal of this assignment is to design and develop a simple 3D scene using Unity 3D that fulfills the following key criteria:

- Incorporates Unity editor-based geometry.
- Includes a camera setup and screenshot capture.
- Uses textures and materials.
- Implements multiple light sources.
- Includes at least one imported asset.
- Features particle effects and animation for realism.

## 1.2 Tools and technologies Used

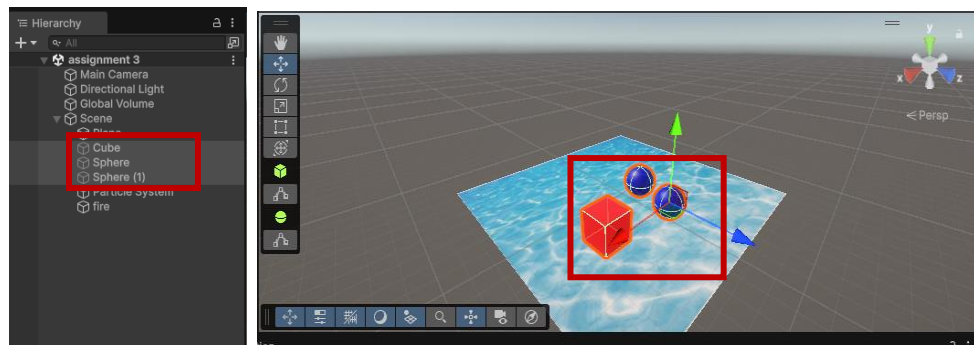
- Game Engine: Unity 6.1
- Platform: Windows
- Assets: Unity default assets, particle system.
- External: Picture to add realism

## 1.3 Scene Components

### 1.3.1 Geometry Added via Unity Editor

**Cube:** Red textured cube placed centrally on the platform.

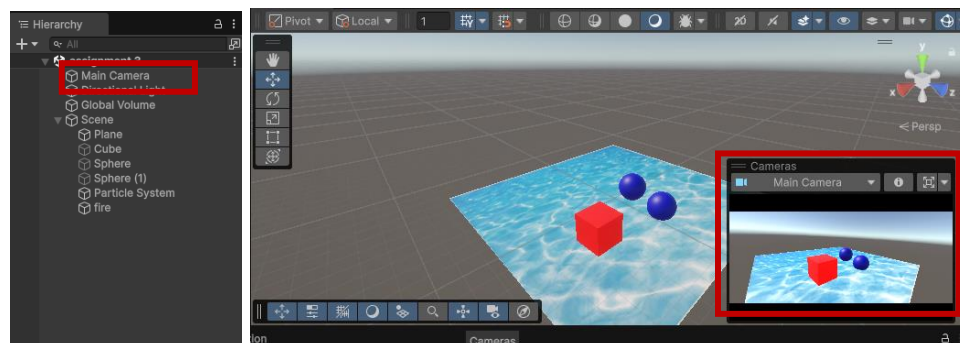
**Spheres (2):** Blue spheres with applied animation and motion.



### 1.3.2 Camera Setup

A **Main Camera** is positioned to capture the main view of the scene.

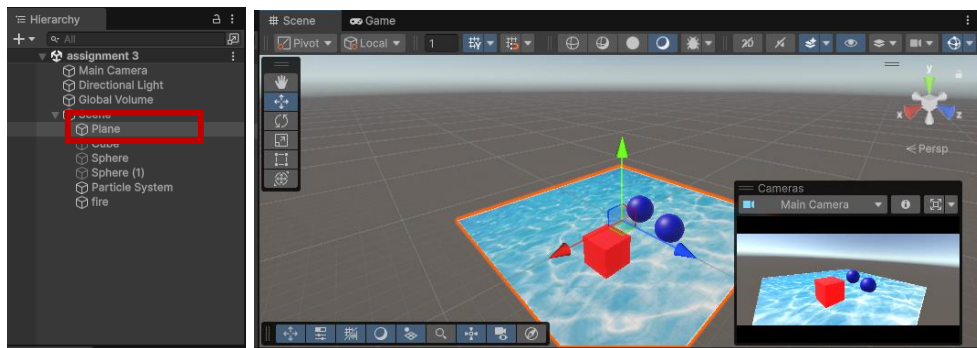
The camera view displays all key elements including the cube, spheres, platform, fire effects, and background.



### 1.3.3 Textures

The platform has a **realistic water texture** applied to it.

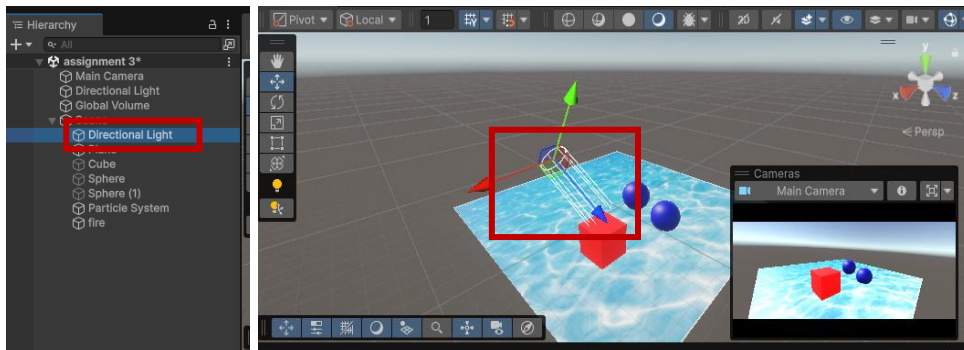
This texture was **imported from an external source** (image file), enhancing the realism of the surface and reflecting light accurately in the scene.



### 1.3.4 Lighting

**Directional Light:** Simulates sunlight for global illumination.

**Particle System Light (Fire/Sparks):** Emulates localized lighting effects like fire and sparks.

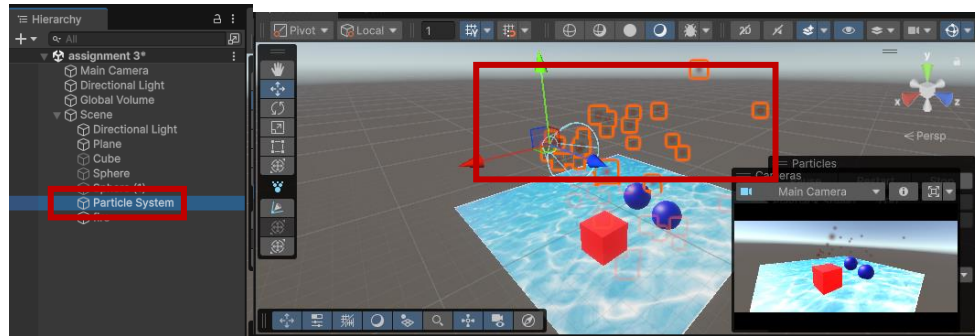


### 1.3.5 Particle System

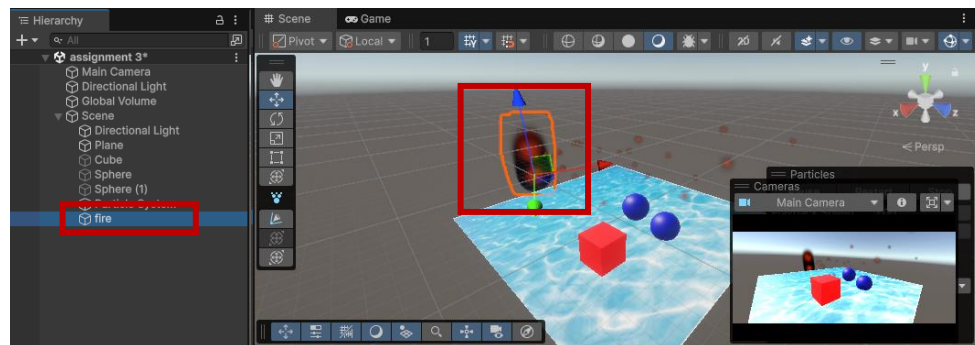
Two particle systems were implemented:

**Sparks:** A red/black particle burst near the background.

**Fire:** Simulates fire, adding ambiance and motion to the scene.



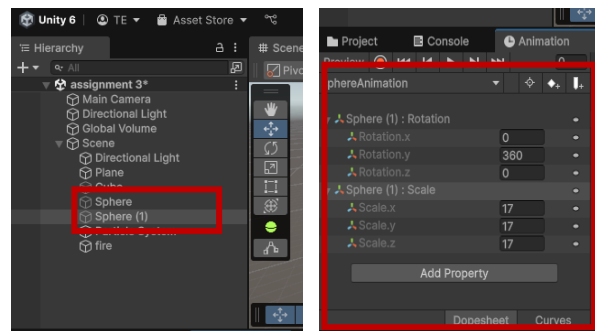
*sparks*



*fire*

### 1.3.6 Animation

Both **spheres** have been animated with motion (rotation) using Unity's Animator or Timeline.



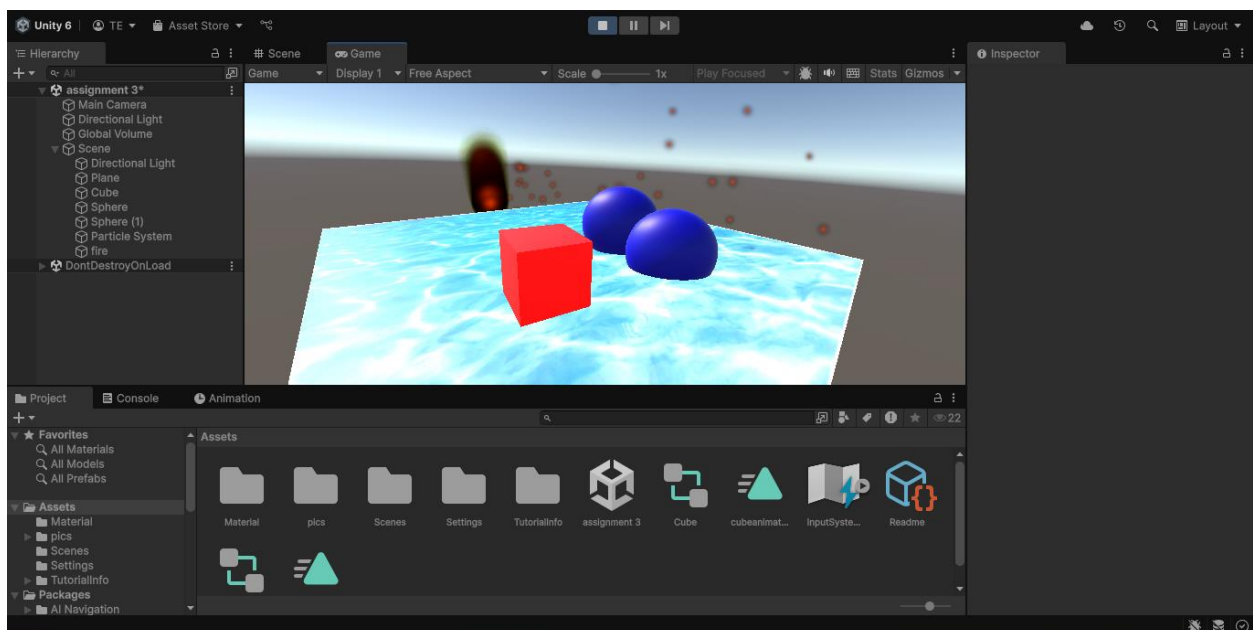
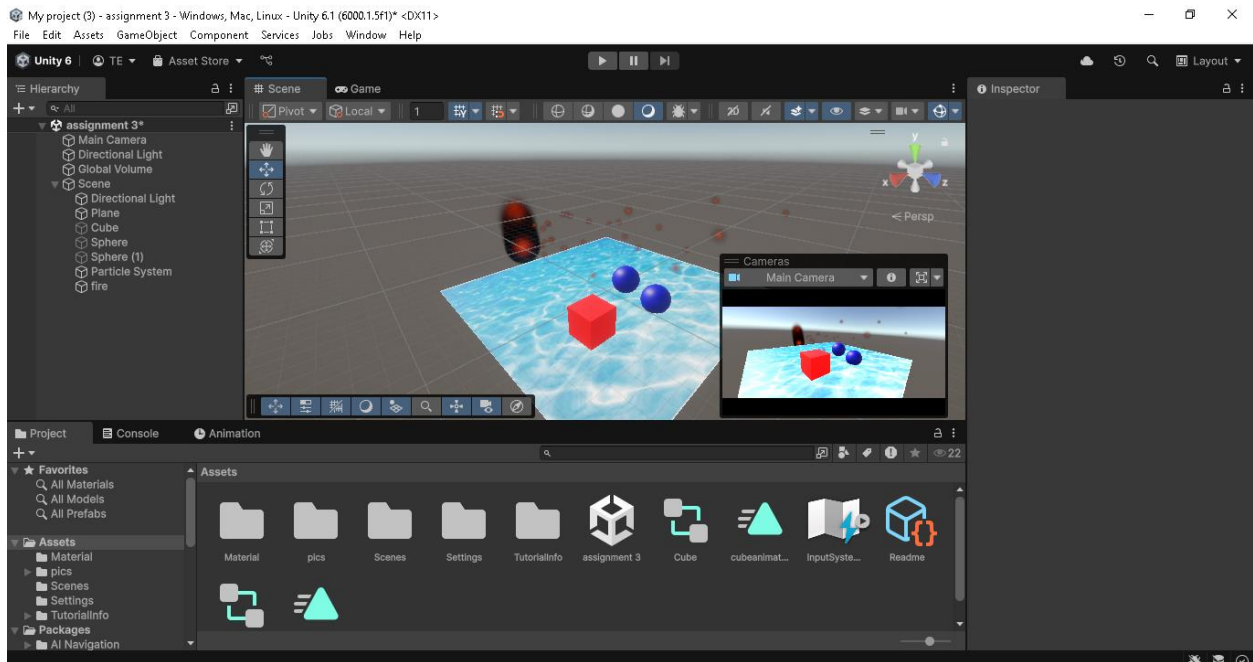
## 1.4 Implementation Details

- GameObjects were created and arranged using the Unity Hierarchy window.
- Custom materials and textures were applied to enhance realism.

- Particle systems were customized for color, size, and looping behavior.
- The scene was saved under the project's assignment 3 directory and backed up regularly.

## 1.5 Screenshots

- Synchronizing particle effects with scene lighting.



## **1.6 Challenges Faced**

- Synchronizing particle effects with scene lighting.
- Adjusting camera to capture optimal scene view.
- Importing and configuring external assets properly.
- Managing animation loop settings for spheres.

## **1.7 Conclusion**

This assignment enabled the practical application of Unity's core functionalities including 3D object placement, animation, lighting, particle systems, and integration of external assets. It provides foundational knowledge required for building more complex and interactive 3D environments.

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